



3D printed or lab-grown organs and prosthetics will prolong our life

sional, two-chamber mouse heart using a modified desktop, inkjet printer. By filling the ink cartridge with cells, they've been able to "publish" functional human kidneys. We're taking technology into places that's truly unlocking the secrets of our physiology like never before.

Just in the last few months, we have witnessed scientific feats like growing human ears on the backs of mice and implant culture-grown lungs into rats. 3D printing ears, lungs, and even tissue samples for use in humans. Thanks to recent advancements in stem cell research, British scientists have successfully created artificial blood that can be administered to anyone, by growing RBCs from fibroblasts that have been reprogrammed into mature red blood cells in the lab. The blood, developed by researchers at the University of Edinburgh and the Scottish National Blood Transfusion Service (SNBTS), would be Type O negative, also known as universal donor blood, which currently comprises just 7 percent of the blood donor pool.

Experts in the field are unanimous in their prediction that in the near future, whenever we need replacement body parts, whether it's a tissue culture or a major organ, we'll just use rejection-proof artificial organs grown in laboratories using our own cells. "By putting in the

parts you need, you'll be able to extend life by several decades," explains Anthony Atala, director of the Wake Forest Institute for Regenerative Medicine. "We may even push that up to 120, 130 years."

Downloading memories

The above mentioned techniques on our path to immortality have overwhelming odds restricted by our physiology. Our body's programmed to degenerate. But is that also true of our mind?

Less than a year ago, Berger and Sam Deadwyler at Wake Forest University successfully conducted experiments in which

"There's this whole other aspect of intelligence which is new thoughts, new ideas, creativity, these are things that we don't even understand fully among us humans yet, much less how to get that into a robot."

- James McLurkin, Asst. Prof of Computer Science, Rice University, USA

they actually inserted memories into the brains of rats by stimulating certain parts of the hippocampus with electrical signals. Berger said they have also been able to disable the hippocampus, in effect blocking the memory, and then electronically stimulating certain areas to create a

"new" memory. This has worked successfully in rats and monkeys, while human trials are yet to begin.

At the Massachusetts Institute of Technology, Ed Boyden leads the Synthetic Neurobiology Group, which is building new tools to explore the brain. Boyden and his colleagues have found a protein in algae that is able to convert light into electricity. When this protein, called channelrhodopsin, is introduced into certain neurons, it's triggered by light to create distinctive patterns they can be translated into electrical impulses and then mapped – resulting in a computer code of a memory. This opens up a whole new range of technologies in the field of brain and memory research, digitizing data and even offloading it to some extent.

The biggest roadblock to downloading memories right now is that they seem to disappear when not in use. Unlike memories in a microchip which can be accessed any time, if the human brain isn't actively triggering a memory, one can't really download or copy it, according to Ted Berger, a neuroscientist at the University of Southern California. And unless we get past this hurdle, the reality of completely offloading the contents of our brain can't be achieved.

Even Stephen Hawking believes that downloading data from the brain is an inevitable eventuality. "I think the brain is like a program in the mind, which is like a computer, so it's theoretically possible to copy the brain onto a computer and so provide a form of life after death," he said, according to a report in Guardian last September. "However, this is way beyond

our present capabilities." But don't be disappointed just yet.

The Avatar Project

This is the most ambitious endeavour towards achieving immortality of the mind, perhaps not the body. And it hints